Console Component

```
                RRR   RA    RB    RC
   PC   INST    ABC  DATA  DATA  DATA   IM
F:0000  0000
NOP

D:0000  0000  000  0000  0000  0000  0000
NOP

E:0000  0000  000  0000  0000  0000  0000
NOP       BWB:0  0000   MWR:0  0000  0000
          RWB:0  0000   MRD:0  0000

M:0000  0000 0
NOP       RWB:0  0000

                 Z:0  N:0  O:0

R0:0 0000 R1:0 0000 R2:0 0000 R3:0 0000
R4:0 0000 R5:0 0000 R6:0 0000 R7:0 0000
```

Console Functionality

- Memory mapped text display
- Visual debugging of the different stages of your pipeline CPU

Memory Mapped Text Display

- 20 x 15 Text display of write only memory
- 1 character is 16 bit word
- Default memory location is 0xFC00 to 0xFDFF
- Top 8 bits are for colour selection when Enable = ‘1’
- Default write enable signal is active high ‘1’
- Write on the rising edge of the clock signal
- Test code for the display is located in

  C:\Dev\assembler\test code\

Character Bit Structure								
15	14	13	12	11	10	9	8	7 down to 0
Colour Enabled	Not Used	Red(1)	Red(0)	Green(1)	Green(0)	Blue(1)	Blue(0)	Character

Colour Output	
Colour Bits	4 bit Colour Output
00	0011
01	0111
10	1011
11	1111

Visual Debugging

- Can display any signal from your pipeline design
- Support bit, nibble, byte and word sizes
- Best used in manual clocking
- Decodes instructions into text description

Adding the Component

To add the component to your project the following is required.

- Signals for VGA, clock and switch added to constraints file.
- Above signals added to top module entity declaration.
- Component declaration and instantiation.
- Text for above available in top portion of console.vhd file.
- Text is commented. To remove comments characters select text and press CTRL /
- Once added and built successfully the display shown in the first slide will be displayed when the bit file is programmed. Make sure the debug console switch is in the upward position.

Display CPU Signals

- To display CPU you are only required to connect the signals in the instantiation statement. Once the project is successfully built again the signals will be displayed.

  s1_pc => fetch_latch_pc,

  s1_inst => fetch_latch_inst,

- Start with the first stage and advance to next one after you have validate that the first is working.